Computer Graphics and Visualization

(Introduction to Computer Graphics and Visualization)

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Week 1

Learning Objectives

By the end of this lecture, students should be able to:

- Define computer graphics and its significance.
- Identify and describe various hardware components used in computer graphics.
- Understand the functionality of different input and output devices.
- Recognize the challenges associated with devices like facsimile machines.
- Identify different types of Computer Graphics.
- Understand basic basic Principles of Animation in Computer Graphics

What is Computer Graphics?

Computer graphics, a dynamic field within computing, involves the creation, manipulation, and rendering of visual content using computers. It includes digital images, animations, and interactive graphics used in various sectors like entertainment, education, scientific visualization, and virtual reality.

Computer graphics can be used in UI design, rendering, geometric objects, and animation. It refers to the manipulation and representation of images or data in a graphical manner, requiring various technology for creation and manipulation, and digital synthesis and its manipulation.

Application of Computer Graphics

Computer graphics user interfaces (GUIs) – A graphic, mouse-oriented paradigm which allows the user to interact with a computer.

Business presentation graphics – "A picture is worth a thousand words".

Cartography – Drawing maps.

Weather Maps – Real-time mapping, symbolic representations.

Satellite Imaging – Geodesic images.

Photo Enhancement – Sharpening blurred photos.

Medical imaging – MRIs, CAT scans, etc. - Non-invasive internal examination.

Engineering drawings – mechanical, electrical, civil, etc. - Replacing the blueprints of the past.

Typography – The use of character images in publishing - replacing the hard type of the past.

Application of Computer Graphics

- **Architecture** – Construction plans, exterior sketches - replacing the blueprints and hand drawings of the past.
- **Art** – Computers provide a new medium for artists.
- **Training** – Flight simulators, computer aided instruction, etc.
- **Entertainment** – Movies and games.
- **Simulation and modeling** – Replacing physical modeling and enactments

Hardware Components in Computer Graphics

A typical computer graphics system comprises several hardware components that work in tandem to produce and display images.

a. Central Processing Unit (CPU): The brain of the computer that performs calculations and processes instructions.

b. Graphics Processing Unit (GPU): Specialized hardware designed to accelerate the rendering of images and animations.

c. Memory

Main Memory (RAM): Stores data and instructions temporarily for quick access.

Frame Buffer: A dedicated portion of memory that holds the image data to be displayed on the screen.

Hardware Components in Computer Graphics

d. Display Devices

Cathode Ray Tube (CRT): An older display technology that uses electron beams to illuminate phosphors on the screen.

Liquid Crystal Display (LCD): A flat-panel display that uses liquid crystals to modulate light.

Light Emitting Diode (LED): An advanced display technology that uses LEDs for backlighting, offering better contrast and energy efficiency.

Input Devices in Computer Graphics

- Input devices allow users to interact with computer graphics systems by providing data and control signals.

a. Keyboard: Primary device for text input and command execution.

b. Mouse: A pointing device that detects two-dimensional motion relative to a surface.

c. Graphic Tablets: Flat surfaces that detect stylus or pen input, allowing for precise drawing and handwriting input.

d. Light Pens: Handheld devices that detect light from the screen, enabling users to point directly at the display to select objects or draw.

Input Devices in Computer Graphics

e. Joysticks: Input devices that detect movement in multiple directions, commonly used in gaming and simulations.

f. Scanners: Devices that convert physical images and documents into digital format.

g. Facsimile (Fax) Machines: Devices that scan documents and transmit them over telephone lines.

Challenges with Facsimile Machines:

- Image Quality: Degradation due to multiple scanning and transmission processes.
- Transmission Errors: Susceptibility to noise and signal loss over telephone lines.
- Compatibility Issues: Variations in standards can lead to incompatibility between devices.

Output Devices in Computer Graphics

☐ Output devices render the processed data into visual or physical forms.

a. Monitors/Displays: Devices that visually present images and graphics generated by the computer.

b. Printers

- Inkjet Printers: Spray tiny droplets of ink directly onto paper.
- Laser Printers: Use laser beams to produce high-quality text and graphics.
- Plotters: Specialized printers used for printing vector graphics, commonly used in engineering and architectural designs.

c. Microfilm: A storage medium that uses photographic film to store documents in miniature form.

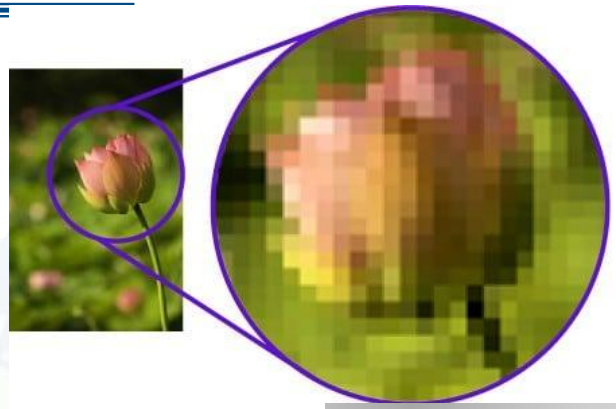
Types of Computer Graphics

Computer graphics can be broadly categorized based on their applications, nature, or the way they are generated. Here are the main types of computer graphics –

- **Raster Graphics (Bitmap Graphics)** – Bitmap graphics are the images made up of tiny dots called pixels (picture elements).
- **Vector Graphics** – Images are created using mathematical equations to represent geometric shapes such as lines, circles, and polygons.
- **3D Graphics** – Graphics that represent three-dimensional objects and scenes, often used for simulations, video games, and movies.
- **Interactive Graphics** – Graphics that allow users to interact with them, typically through a user interface (UI).
- **Real-Time Graphics** – Graphics that are rendered in real-time, meaning they are created and displayed instantly as the user interacts with them.

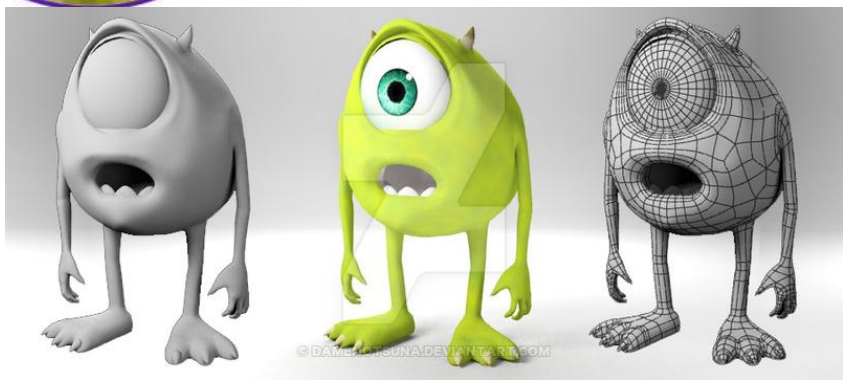
Types of Computer Graphics

- ☐ Raster Graphics (Bitmap Graphics)

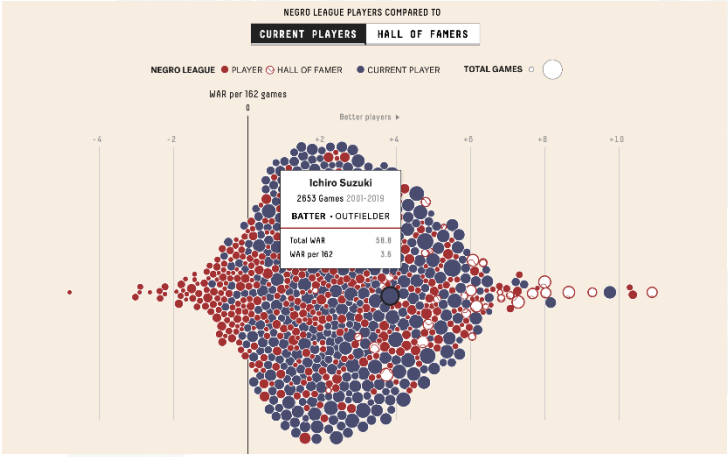


- ☐ Vector Graphics

- ☐ 3D Graphics



- ☐ Interactive Graphics



- ☐ Real-Time Graphics

Basic Principles of Animation in Computer Graphics

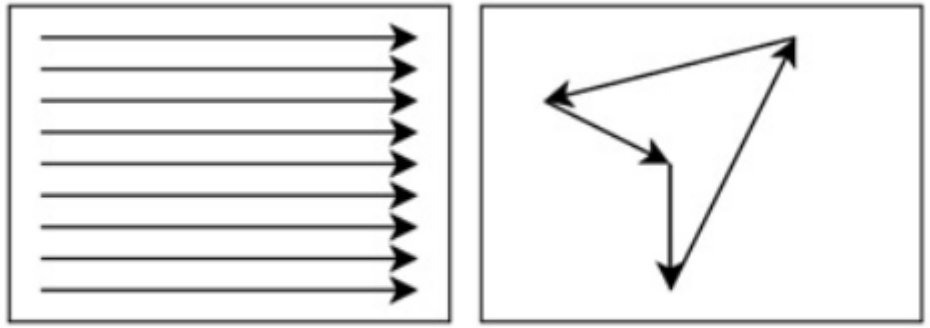
- The animation in computer graphics is nothing but a sequence of frames where each frame is generating a graphics output on the scene. Generally, not all frames are generated in a similar way.

- The concept of **interpolation** is used for animation creation. These can be done with linear, quadratic or other interpolation types. There are several types of animation including **key-framing**, **morphing**, **tween animations**, etc.

Raster and Vector Graphics

- Raster and vector graphics are two common types of digital images.

- Raster graphics consist of tiny pixels; they are ideal for detailed and colourful images like photographs. On the other hand, vector graphics use mathematical paths for scaled designs like logos and illustrations.

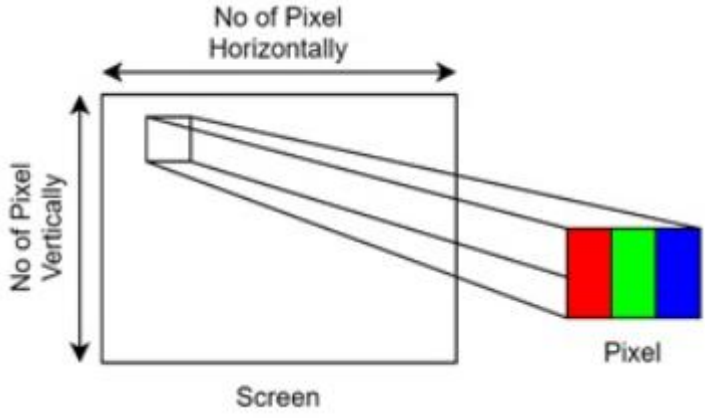


Raster Vector

- Raster images use bit maps to store information, requiring a larger bitmap for larger files. Vector graphics, on the other hand, use sequential commands or mathematical statements to place lines or shapes in a 2-D or 3-D environment.

Pixel and Image Resolution

- Pixels, derived from the term "picture element," are the fundamental unit of digital images, representing visual information. They consist of RGB subpixels, which mix to create diverse colors. Pixels form a raster image, with each pixel representing a specific position and color value.



- In an image, how many pixels are there in row and in column manner that signifies the resolution. An image with large number of pixels inside it will be more accurate and detailed image as compared to lower number of picture elements.

What is Rendering in Computer Graphics?1

- The term rendering states visualizing the data. In computer graphics showing the generated graphics is rendering. However we generally use 3D rendering with the term "rendering".

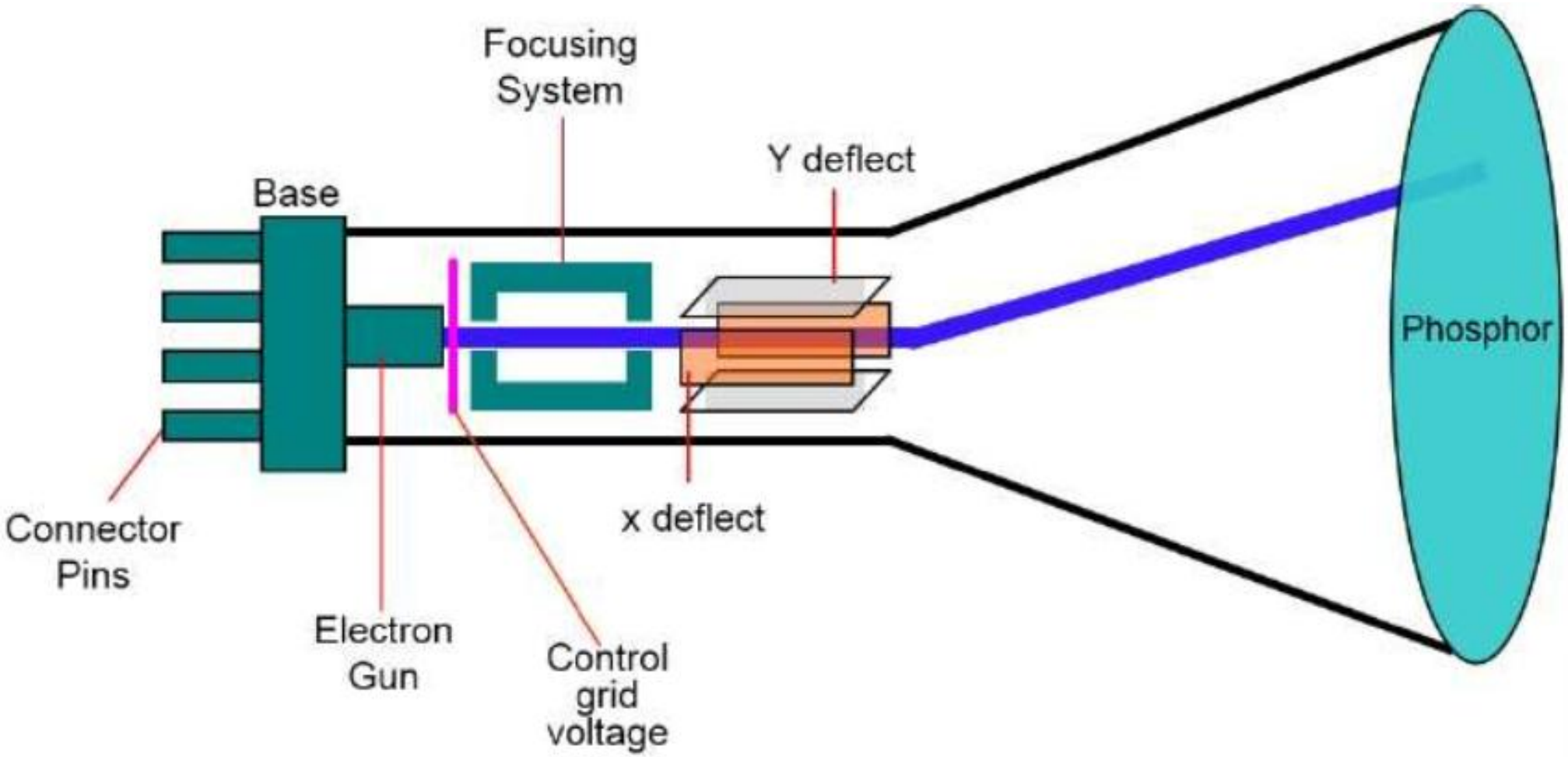
- 3D rendering is a computer-based process that generates a 2D image from a digital 3D scene. It is essential in various industries like architecture, product design, advertising, video games, and visual effects.

What is Rendering in Computer Graphics?2

- Some of the rendering techniques are listed below –

- Scanline is a traditional rendering technique that calculates computer graphics surface.
- Z-Buffer is a two-dimensional data system used for depth calculation and storage.
- Shading and lighting manipulate light and dark levels using software effects.
- Texture / Bump Mapping displays color, material, and object details.
- Ray Tracing and Ray Casting are useful rendering techniques for creating natural lighting effects.

Cathode Ray Tube



Cathode Ray Tube

- The primary output device in a graphical system is the video monitor. The main element of a video monitor is the Cathode Ray Tube (CRT), shown in the following illustration.

- The operation of CRT is very simple –

  - The electron gun emits a beam of electrons (cathode rays).
  - The electron beam passes through focusing and deflection systems that direct it towards specified positions on the phosphor-coated screen.
  - When the beam hits the screen, the phosphor emits a small spot of light at each position contacted by the electron beam.
  - It redraws the picture by directing the electron beam back over the same screen points quickly.

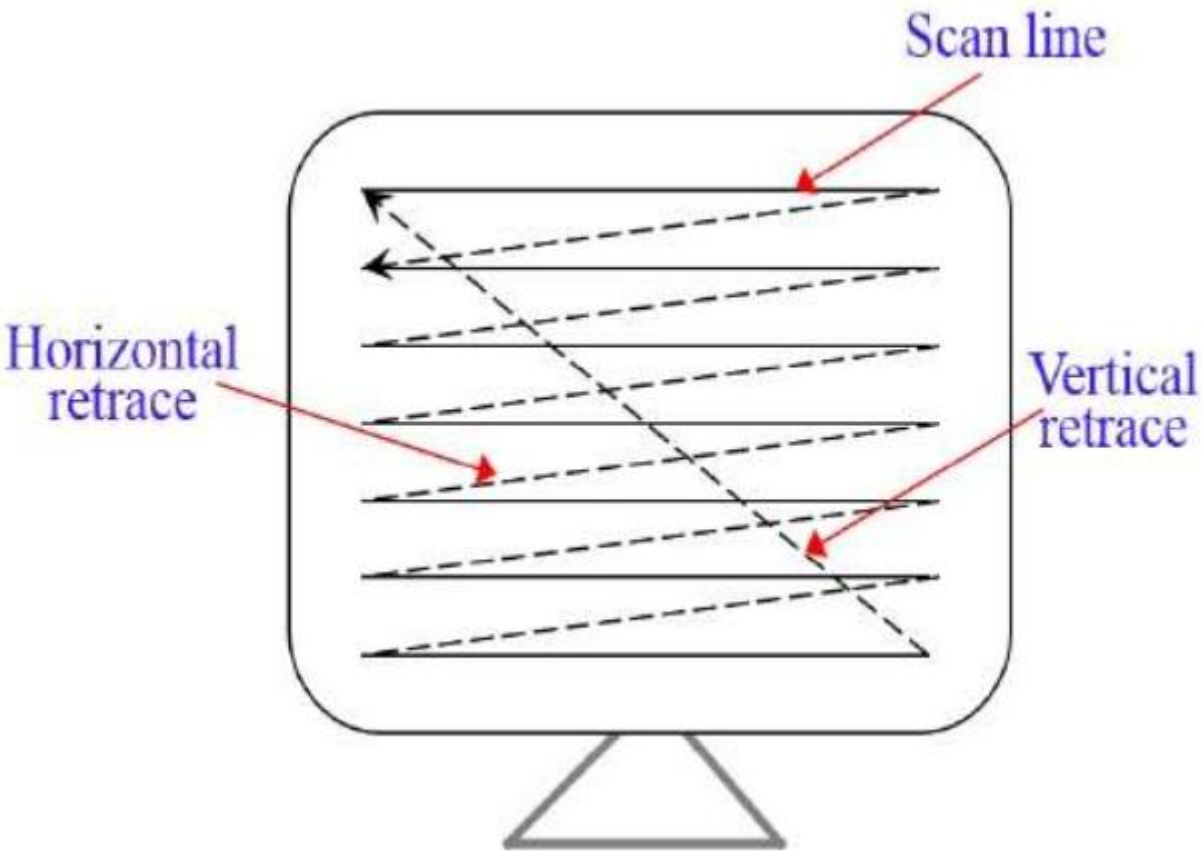
Raster Scan

- In a raster scan system, the electron beam is swept across the screen, one row at a time from top to bottom. As the electron beam moves across each row, the beam intensity is turned on and off to create a pattern of illuminated spots.

- Picture definition is stored in memory area called the **Refresh Buffer** or **Frame Buffer**. This memory area holds the set of intensity values for all the screen points. Stored intensity values are then retrieved from the refresh buffer and painted on the screen one row (scan line) at a time as shown in the following illustration.

- Each screen point is referred to as a **pixel (picture element)** or **pel**. At the end of each scan line, the electron beam returns to the left side of the screen to begin displaying the next scan line.

Raster Scan



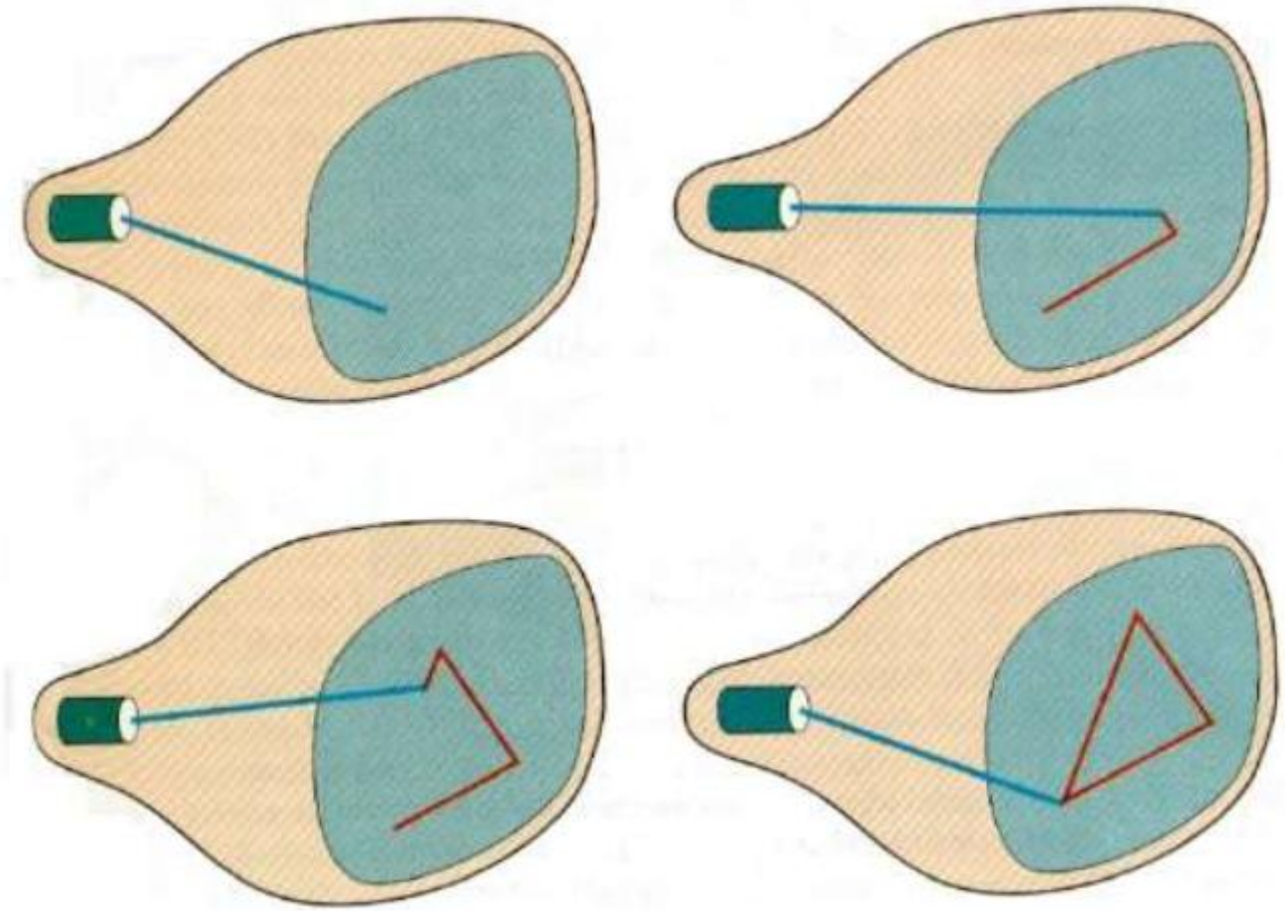
Random Scan (Vector Scan)

- In this technique, the electron beam is directed only to the part of the screen where the picture is to be drawn rather than scanning from left to right and top to bottom as in raster scan. It is also called vector display, stroke-writing display, or calligraphic display.

- Picture definition is stored as a set of line-drawing commands in an area of memory referred to as the refresh display file. To display a specified picture, the system cycles through the set of commands in the display file, drawing each component line in turn. After all the line-drawing commands are processed, the system cycles back to the first line command in the list.

- Random-scan displays are designed to draw all the component lines of a picture 30 to 60 times each second.

Random Scan (Vector Scan)



Further Reading 1

- ☐ How do display devices like CRT and LCD work in the context of graphics?
- ☐ What is the role of linear algebra in computer graphics?
- ☐ What is a Frame Buffer?
- ☐ How do Spatial Data Structures improve rendering efficiency?
- ☐ What is the significance of Color Depth?
- ☐ What is anti-aliasing, and why is it important?
- ☐ What is the difference between 2D and 3D graphics?
- ☐ What are Homogeneous Coordinates?
- ☐ What is Graphics Pipeline?
- ☐ What are Transformation Matrices in computer graphics?
- ☐ How do Translation, Scaling, and Rotation work in 2D / 3D transformations?
- ☐ What is a View Matrix in graphics rendering?
- ☐ What is Perspective Projection? How is it different from Orthographic Projection?

Further Reading 2

- ☐ What is the concept of Clipping in graphics?
- ☐ What are vectors and how are they used in computer graphics?
- ☐ What is Ray Tracing, and how does it differ from Rasterization?
- ☐ What is the role of normal vectors in lighting calculations?
- ☐ What is the significance of Depth Buffers in 3D rendering?
- ☐ What is Bresenham's line-drawing algorithm?
- ☐ What is Scanline Algorithm, and how is it used in raster graphics?
- ☐ What is Midpoint Circle Algorithm?
- ☐ How does the Flood-Fill Algorithm work?
- ☐ What is Texture Mapping?
- ☐ What is the purpose of Z-buffer in 3D rendering?
- ☐ How do graphics techniques for VR differ from traditional 3D graphics?
- ☐ What are some common optimization techniques used in real-time graphics?

THANK YOU FOR LISTENING